

Following completion of the preparation materials, students should be able to:

1. Compare and contrast the within-class and between-class pharmacokinetic properties of the antiretroviral agents.
2. Describe the structure-activity relationships, mechanism of action, and mechanisms of viral resistance of each class of antiretroviral agents in relation to specifically targeted viral proteins and give examples.
3. Explain the clinical rationales for selection of specific antiretroviral therapy (ART) and give examples for:

ART-naïve patients	Transgender patients
Switch to 2-drug ART	Women (assigned female at birth)
Treatment-experienced patients	Pregnant people and peripartum care
Pre-exposure prophylaxis (PrEP)	Neonates
Post-exposure prophylaxis (PEP)	Pediatric patients
	Older patients
4. Differentiate the class and drug-specific adverse effects and their impact on patient adherence to therapy.
5. Explain the mechanism-related and metabolism- and transport-related drug interactions for specific drugs, including the antiretroviral drugs that should not be used in combination.

What you need to know and understand

- HIV life cycle → Drug targets

(1) Entry (binding and fusion to host cell), (2) reverse transcription, (3) integration of HIV cDNA into host genome, (4) replication (transcription and translation), (5) assembly and budding (6) maturation

- ART resistance: HIV reverse transcriptase is error prone and lacks a proofreading function. Mutation is frequent. Drug therapy provides the necessary *selective pressure* to promote growth of drug-resistant viruses that arise naturally. Resistance testing is important to help guide treatment decisions in both treatment-naïve and treatment-experienced patients.
- Goal of therapy: To restore immunocompetence – increase the CD4 cell count and reduce viral load to undetectable (suppression of HIV RNA replication), reduce HIV-associated morbidity and prolong the duration and quality of survival, and prevent HIV transmission
- Drug classes: nucleoside reverse transcriptase inhibitors (NRTIs), integrase strand transfer inhibitors (INSTIs), nonnucleoside/tide reverse transcriptase inhibitors (NNRTIs), protease inhibitors (PIs), entry inhibitors, and capsid inhibitor.
- Pharmacokinetic enhancers: Two pharmacokinetic enhancers (“boosters”), ritonavir and cobicistat, are potent inhibitors of CYP3A4, which increases systemic levels of the partner drug allowing for less frequent dosing and less variation in drug levels, resulting in enhanced efficacy and reduced resistance.

- ART principles apply across all populations.
- Recommended initial therapy in treatment-naïve patients consists of a 2 NRTI backbone with an integrase inhibitor (INSTI-based ART). An NNRTI or boosted PI backbone may be considered in certain clinical situations.
- Note: Several 2-drug combination regimens are FDA approved for stable patients who meet specific criteria.
- Treatment-experienced patients require a new regimen based on evaluation of the patient's past ART and previous and current genotype and/or phenotype of the virus. Drug resistance tests determine which drugs are likely to be fully or partially active.
- NRTIs
 - First-generation – didanosine, stavudine, zidovudine: Most toxicities are due to inhibition of mitochondrial DNA polymerase causing peripheral neuropathy, pancreatitis, and lipoatrophy and life-threatening hepatotoxicity characterized by lactic acidosis, hepatomegaly, and steatosis. Of the first-generation drugs, only zidovudine is available in the US.
 - Second-generation – tenofovir (TDF and TAF), emtricitabine, lamivudine, and abacavir. TDF can cause proximal renal tubulopathy and bone loss but TAF has less long-term kidney and bone toxicity. Lamivudine and emtricitabine are structurally similar and should not be used together. Abacavir can cause a life-threatening hypersensitivity reaction. Patients should be screened for HLA-B*5701 allele.

What you need to know and understand – Adverse Effects

- INSTIs:
 - Integrase inhibitors are bictegravir, dolutegravir, elvitegravir, and raltegravir. They are well tolerated, highly effective, and have a high barrier to resistance. They are recommended first line drugs with 2NRTIs for initial ART.
- NNRTIs:
 - This group is associated with varying degrees of GI intolerance and skin rash.
 - The two most frequently used agents, efavirenz and rilpivirine, cause QT interval prolongation and are associated with neurological and psychiatric side effects.
 - NNRTIs are metabolized by, and are inhibitors and inducers of, various CYPs. There is a high potential for drug-drug interactions.
- PIs:
 - Class side effects include insulin resistance, hyperglycemia, diabetes, hyperlipidemia, lipodystrophy, hepatotoxicity, bleeding in patients with hemophilia, and PR interval prolongation. PIs are CYP3A4 substrates and are used with boosters – CYP3A4 inhibitors. There is a high potential for drug-drug interactions.

What you need to know and understand – adverse effects

- **EIs: Entry inhibitors** have different mechanisms of action, pharmacokinetic profiles, clinical uses, and adverse effects.
 - **CCR5 antagonists:** Maraviroc is generally well tolerated. The drug has been associated with severe hepatotoxicity and related rash or fever. Liver function monitoring is recommended.
 - **Fusion inhibitor:** Enfuvirtide is a peptide given by subcutaneous injection twice daily and often causes local cutaneous reactions.
 - **Attachment inhibitor:** Fostemsavir is an oral prodrug (active temsavir). Its most common side effect is nausea. Other effects are QT interval prolongation and elevated liver enzymes in patients with hepatitis B or hepatitis C viral infections. Strong CYP3A4 inducers should be avoided.
 - **Post attachment inhibitor:** Ibalizumab is given by intravenous infusion every 2 weeks. The most common adverse effects are diarrhea, dizziness, nausea, and rash.
- **Capsid inhibitor:** Lenacapavir directly binds the interface between capsid protein (p24) subunits in hexamers, changing its conformation, which stabilizes it, prevents viral uncoating, and disrupts the viral lifecycle at several steps.

Viral Resistance to Antiretroviral Drugs

HIV reverse transcriptase is error prone and lacks a proofreading function → mutation is frequent

- HIV has a high mutation rate and a tremendous number of infectious virions.

Transmitted drug resistance is acquired through person to person spread.

Virologic failure:
Inability to achieve or maintain suppression of viral replication to an HIV RNA level <200 copies/mL.

There is a high likelihood that all untreated infected individuals harbor viruses with single amino acid mutations that confer some degree of resistance.

Principles of ART

ART has dramatically reduced HIV-associated morbidity and mortality and has transformed HIV infection into a manageable chronic condition, with life expectancy approaching that for people without HIV.

Current standard of care is 3-4 antiretroviral agents for entire duration of treatment.

2 NRTIs + an INSTI, or an NNRTI, or a boosted PI

Preferred initial ARV regimen

Some 2-drug regimens are FDA approved in selected patients.

Outcome of initial therapy should be:

- Undetectable viral load (plasma HIV RNA <20-75 copies/mL) within 24 weeks of starting treatment.
- CD4 count increase of 50-150 cells/mm³ in first year

Treatment failure generally requires implementation of a completely new combination of drugs.

It is usually a consequence of nonadherence.

NRTI: Nucleoside reverse transcriptase inhibitor; NNRTI: Nonnucleoside reverse transcriptase inhibitor;
PI: Protease inhibitor; INST: Integrase strand transfer inhibitor

Adverse Effects of Antiretroviral Therapy

Immune reconstitution inflammatory syndrome:

- reversal of immunodeficiency → ↑ host immune system → inflammatory reaction to overt or subclinical opportunistic infections or malignancies

Class- and drug-specific toxicities:

- Later-generation drugs are generally more tolerable than the first-generation versions.

Significant pharmacokinetic drug interactions:

- substrates, inhibitors and/or inducers of metabolic enzymes and transporters

Viral Resistance to NRTIs: Point Mutations

- Usually, high level resistance requires 3 to 4 codon substitutions.
- However, **M184V/I** single amino acid substitution
 - Confers high-level resistance to **lamivudine and emtricitabine**
 - Confers low-level resistance to **abacavir**
 - But restores susceptibility to **tenofovir and zidovudine**.
- Other common mutations are K65R and L74R
- Cross-resistance is reported following prolonged therapy.

M=methionine, V=valine, I=isoleucine, K=lysine, R=arginine, L=leucine

M184V/I single mutation of methionine-to-valine or isoleucine substitution at position 184 position.

Features of Specific NRTIs: Tenofovir DF (TDF) and Tenofovir AF (TAF)

Adenosine analogs.

Oral, well absorbed, P-glycoprotein substrate, minimal hepatic metabolism, urine as tenofovir, $t_{1/2\text{intracellular}}$ 10-50 h

TDF is rapidly converted after absorption by hydrolases to tenofovir, which is taken up by host cells and metabolized to TFV-DF.

TAF is administered at a lower dose than TDF → lower plasma concentrations.

However, intracellular concentrations of active tenofovir diphosphate are higher.

TAF circulates as uncleaved prodrug, which is taken up into cells. TAF is then metabolized intracellularly to TFV-DF.

TDF and TAF have broad therapeutic applications and are available in numerous fixed-dose combinations.

- First-line NRTIs for treatment of HIV-1
- Pregnant patients, all trimesters
- TDF Children ≥ 2 years old
- TAF Children, adolescents ≥ 12 years old
- Pre-exposure prophylaxis (PrEP)
- Post-exposure prophylaxis (PEP)
- Chronic **HBV** treatment and prophylaxis in immunocompromised patients

Both TDF and TAF suppress HBV infection → Severe acute exacerbation of HBV infection and hepatitis can occur in HIV/HBV coinfecting patients treated with TDF or TAF if the NRTI is discontinued. (US Boxed Warning)

TDF and TAF Adverse Effects and Drug Interactions

TDF and TAF are generally well tolerated.

- **TAF** has lower plasma concentrations and has less long-term kidney and bone toxicity compared with TDF.

Main toxicities: Kidney and bone adverse effects

- **Renal impairment:** Increases in serum creatinine, proteinuria, glycosuria, hypophosphatemia, acute tubular necrosis
- **Kidney injury:** Proximal tubular dysfunction (Fanconi syndrome), acute kidney injury, chronic kidney disease, nephrogenic diabetes insipidus
 - Monitor renal function. Avoid TDF (if possible) in patients with baseline risk factors for kidney injury.
- **Bone loss:** Decreased bone mineral density

Drug interactions:

- Tenofovir + abacavir → high rates of virologic failure
- P-glycoprotein inhibitors: Ritonavir or cobicistat boosted drugs may increase tenofovir levels.
- TAF only – P-gp inducers are contraindicated: Rifampin, carbamazepine, oxcarbazepine, phenytoin, phenobarbital ↓TAF levels → ↓ efficacy

NRTIs Drug-specific properties: Lamivudine and Emtricitabine

Cytidine analogs. Emtricitabine is fluorinated lamivudine. *They should not be used together.*

Oral, well absorbed	$t_{1/2}^{\text{intracellular}}$
Good bioavailability	lamivudine 10-15 h (high dose for 1x daily dosing),
excreted in urine as unchanged drug	emtricitabine ~40 h (once daily)

- Both drugs suppress HBV infection

Severe acute exacerbation of HBV infection and hepatitis can occur in HIV/HBV coinfecting patients treated with lamivudine or emtricitabine if the NRTI is discontinued. (US Boxed Warning)

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- Resistance: M184V/I point mutations → high level resistance.
 - K65R, L74R also reduce susceptibility.
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Well tolerated

Emtricitabine may cause hyperpigmentation, usually on the hands and/or soles of the feet

Features of Specific NRTIs: Abacavir and Zidovudine

Abacavir guanosine analog

Oral, well absorbed, hepatic metabolism by alcohol dehydrogenase, glucuronyl transferase, urine as metabolites, $t_{1/2\text{intracellular}}$ 12-26 h
Intracellular metab.: abacavir monophosphate → carbovir MP →→ active carbovir TP

Pharmacogenomics: All patients must be screened for HLA-B*5701

- Hypersensitivity reactions: multiorgan involvement, *potentially fatal*

Abacavir should not be given to patients with the HLA-B*5701 genotype nor restarted in patients who have a hypersensitivity reaction.

- Resistance: M184V/I and K65R frequently found in patients receiving abacavir
- Frequently paired with lamivudine as a component of NRTI backbone
- Avoid using with tenofovir due to high rate of virologic failure with the combination
- Alcohol (EtOH) → ↑ abacavir levels

Zidovudine thymidine analog

Oral (2x daily), IV (continuous infusion), hepatic glucuronidation, urine unchanged drug, $t_{1/2\text{intracellular}}$ ~7 h

Therapeutic uses:

- Perinatal IV infusion to prevent vertical transmission
- ART for neonates

Zidovudine is otherwise rarely used: twice daily dosing, more severe adverse effects, and inferior virologic potency (zidovudine-lamivudine v. TDF-emtricitabine ART)

Questions

1. What are the important (clinically relevant) take-aways about the PK properties of the NRTIs class?
2. How is viral resistance related to NRTIs' mechanism of action and effects?
3. How is mitochondrial toxicity related to the NRTIs' mechanism of action?

Questions

1. What are the main toxicities of tenofovir and abacavir.
2. Which of the second-generation NRTIs should not be combined and why?
3. What screening test must be performed prior to administering abacavir and why?
4. What drugs can cause reactivation of HBV and hepatitis when withdrawn in patients with HIV/HBV coinfection?

INSTIs Therapeutic Uses and Resistance

Activity (as a class)

- Potent against both HIV-1 and HIV-2
- Treatment-naïve and treatment-experienced
 - Adults
 - Adolescents, and
 - Pediatrics

The INSTIs are generally well tolerated.

Headache and GI effects are the most frequently reported side effects.

Cabotegravir: Injection site reactions

Resistance

- Primary mutations in integrase gene (Q184R/H/K, or N155H) and
- Additional secondary mutations can accumulate and cause high-level resistance and cross-resistance.

Dolutegravir and bictegravir have a high barrier to resistance.

Raltegravir and elvitegravir have a lower barrier to resistance.

Cabotegravir ART: long-acting IM suspension

Cabotegravir is available in oral form, as a long-acting IM suspension for injection and fixed-dose cabotegravir and rilpivirine long-acting suspension for injection.

For gluteal administration only (may not be an option for people with buttocks implants or fillers).

1. Treatment: Cabotegravir + rilpivirine

- Oral lead-in (optional, to determine tolerability):
daily oral cabotegravir + daily oral rilpivirine x 1 month
 - Continuation: IM cabotegravir-rilpivirine every 4 to 8 weeks (dosing differs)
- Oral bridging therapy: Patients who plan to miss a scheduled dose of IM CAB-RPV by >7 days.

Required criteria: HIV-1 infections in adults and adolescents...

Who have no baseline resistance to either medication,

Who have no prior virologic failures,

Who do not have active hepatitis B virus (HBV) infection (unless also receiving an oral HBV active regimen),

Who are not pregnant and are not planning on becoming pregnant, *and*

Who are not receiving medications with significant drug interactions with oral or injectable CAB or RPV.

2. PrEP: HIV-negative individuals, IM cabotegravir with or without oral lead-in

Questions

1. What properties make INSTIs the preferred first-line choice for ART backbone?
2. Which two are the first-line INSTIs?
3. What is the role of cabotegravir-rilpiverine 2-drug ART in the treatment of HIV-1 patients?
4. What is another therapeutic use of cabotegravir IM?

- Antiretroviral therapy (ART) consists of a combination of antiretroviral drugs that suppress HIV replication. The combination increases potency against the virus and reduces the likelihood of the virus developing resistance. ART reduces the morbidity and mortality rates and improves quality of life. Suppressing viral replication below a detectable level reduces transmission. ART is recommended for everyone who has HIV. Adherence to therapy is essential for suppression of replication. The patient's barriers to adherence should be assessed prior to therapy and at every visit.
- Drugs targeting every step in the HIV replication cycle have been developed. The four classes of drugs for initial treatment are NRTIs, INSTIs, NNRTIs, and PIs.
- Resistance arises from point mutations in binding sites. Transmitted drug resistance is acquired through person to person spread. There is a high likelihood that all untreated infected individuals harbor viruses with single amino acid mutations that confer some degree of resistance.

- Two second-generation NRTIs form the ART backbone for initial therapy: tenofovir disoproxil fumarate, tenofovir alafenamide, emtricitabine, lamivudine, and abacavir. Lamivudine and emtricitabine are structurally similar and not used together. Tenofovir and abacavir should not be used together because of high rates of virologic failure.
- NRTIs have a low affinity for host DNA polymerase except mitochondrial DNA polymerase-gamma.
- Many NRTI toxicities are believed to be due to mitochondrial toxicity: myopathy, peripheral neuropathy, pancreatitis, lipoatrophy, and hepatic steatosis, and lactic acidosis, which can be fatal. Pancreatitis, bone marrow suppression, and cardiomyopathy may be related to mitochondrial toxicity. First-generation NRTIs are most associated with mitochondrial toxicities. Tenofovir, TDF, can cause proximal renal tubulopathy and bone mineral loss. TAF is administered in lower doses and has less kidney and bone toxicity than TDF. Abacavir can cause life-threatening hypersensitivity reaction. Screening for HLA-B*5701 allele is necessary prior to starting therapy.
- NRTIs are oral and excreted in the urine. Abacavir is metabolized by alcohol dehydrogenase and glucuronyl transferase.

- Integrase strand transfer inhibitors (INSTIs) are bictegravir, dolutegravir, elvitegravir (boosted with cobicistat), raltegravir, and cabotegravir. BIC and DOL are first-line ART backbone with 2 NRTIs for treatment-naïve patients. Cabotegravir long-acting IM suspension is a PrEP option and is formulated with rilpivirine as 2-drug ART for treatment of selected, stable HIV-1 patients. CAB is formulated as a tablet for oral lead-in or bridging therapy.
- INSTIs inhibit the insertion of proviral DNA into the host cell genome.
- Resistance occurs with single point mutations in the integrase gene. BIC and DOL have a high barrier to resistance. RAL and ELV have a lower barrier to resistance.
- The INSTIs are taken orally once daily (raltegravir twice daily or once daily at a higher dose). DOL, RAL, and CAB are UGT1A1 substrates. BIC and ELV are CYP3A4 substrates. Efficacy can be reduced when administered with UGT1A1, CYP3A4, and P-glycoprotein inducers.
- The INSTIs are well tolerated. Headache and GI discomfort are the most frequently reported side effects.